MEDIUM

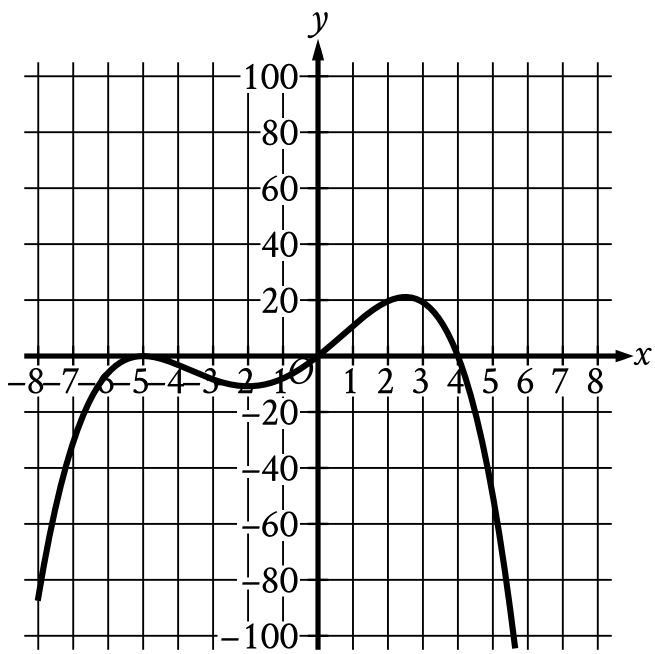
ADVANCED MATH

Nonlinear functions

02

10 questions · ~15 min

Q1 ADVANCED MATH



- In quadrant 3:
  - The curve rises sharply to touch the x axis at point (negative 5 comma 0).
  - The curve falls gradually to a relative minimum at point (negative 2 comma negative 11).
  - The curve rises gradually to cross both axes at the origin.
- In quadrant 1:
  - The curve rises gradually to a relative maximum at point (2.5 comma 21).
  - The curve falls sharply to cross the x axis at point (4 comma 0).
- In quadrant 4 the curve falls sharply.

Which of the following could be the equation of the graph shown in the *xy*-plane?

A. $y = -\frac{1}{10}xx - 4x + 5$

B. $y = -\frac{1}{10}xx - 4x + 5^2$

C. $y = -\frac{1}{10}xx - 5x + 4$

D. $y = -\frac{1}{10}xx - 5^2x + 4$

Q2

ADVANCED MATH

A certain college had 3,000 students enrolled in 2015. The college predicts that after 2015, the number of students enrolled each year will be 2% less than the number of students enrolled the year before. Which of the following functions models the relationship between the number of students enrolled, $f(x)$, and the number of years after 2015, x ?

A. $f(x) = 0.02(3{,}000)^x$

B. $f(x) = 0.98(3{,}000)^x$

C. $f(x) = 3{,}000(0.02)^x$

D. $f(x) = 3{,}000(0.98)^x$

Q3

ADVANCED MATH

$$gx = x^2 + 55$$

What is the minimum value of the given function?

A. 0

B. 55

C. 110

D. 3,025

Q4

ADVANCED MATH

An object is kicked from a platform. The equation $h = -4.9t^2 + 7t + 9$ represents this situation, where h is the height of the object above the ground, in meters, t seconds after it is kicked. Which number represents the height, in meters, from which the object was kicked?

A. 0

B. 4.9

C. 7

D. 9

Q5 ADVANCED MATH

$$px + 57 = x^2$$

The given equation relates the value of x and its corresponding value of px for the function p. What is the minimum value of the function p?

A. -3,249

B. -57

C. 57

D. 3,249

Q6 ADVANCED MATH

The function f is defined by $f x = 2700.1^{x}$. What is the value of $f 0$?

A. 0

B. 1

C. 27

D. 270

Q7 ADVANCED MATH

The function $f(x) = 206(1.034)^x$ models the value, in dollars, of a certain bank account by the end of each year from 1957 through 1972, where x is the number of years after 1957. Which of the following is the best interpretation of "$f(5)$ is approximately equal to 243" in this context?

A. The value of the bank account is estimated to be approximately 5 dollars greater in 1962 than in 1957.

B. The value of the bank account is estimated to be approximately 243 dollars in 1962.

C. The value, in dollars, of the bank account is estimated to be approximately 5 times greater in 1962 than in 1957.

D. The value of the bank account is estimated to increase by approximately 243 dollars every 5 years between 1957 and 1972.

Q8 ADVANCED MATH

x	fx
-1	10
0	14
1	20

For the quadratic function f, the table shows three values of x and their corresponding values of fx. Which equation defines f?

A. $fx = 3x^2 + 3x + 14$

B. $fx = 5x^2 + x + 14$

C. $fx = 9x^2 - x + 14$

D. $fx = x^2 + 5x + 14$

Q9 ADVANCED MATH

$$gx = 11\frac{1}{12}^{x}$$

If the given function g is graphed in the xy-plane, where $y = gx$, what is the y-intercept of the graph?

A. $0, 11$

B. $0, 132$

C. $0, 1$

D. $0, 12$

Q10 ADVANCED MATH

$$f x = 3{,}0000.75^{x}$$

A conservation scientist implemented a program to reduce the population of a certain species in an area. The given function estimates this species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of 3,000 in this context?

A. The estimated percent decrease in the population for this species and area every 8 years after 2008

B. The estimated percent decrease in the population for this species and area each year after 2008

C. The estimated population for this species and area 8 years after 2008

D. The estimated initial population for this species and area in 2008